Angad Singh

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SKILLS

Languages : Python, C, Java, JavaScript, TypeScript, SQL, HTML/CSS
Frameworks & Libraries : Next.js, React, FastAPI, Streamlit, Tailwind CSS, Framer Motion, Pandas, NumPy, Matplotlib
Tools & Platforms : Git, GitHub, VS Code, SQLite, Docker, PostgreSQL, Redis, Pytest
Concepts : Object-Oriented Programming (OOP), REST APIs, Data Structures & Algorithms
Cloud / AI / Data : AWS (EC2, S3), OpenAI API, Google Gemini API, WebSockets, SSE, Prompt Design, Data Validation

EXPERIENCE

AWS Cloud Club (McMaster University) Nov 2025 – Present
Cloud Engineer Hamilton, ON
- Selected as a core team member to help build and scale a student cloud computing community focused on real-world technical skill development
- Collaborating with a cross-functional leadership team to design and deliver hands-on workshops, technical talks, cloud labs, and project nights
- Contributing to student-led, cloud-powered projects using industry tools and collaborative workflows (GitHub, Slack, Notion)

Garth Webb Secondary School Sep 2024 – Feb 2025
Mathematics Tutor Oakville, ON
- Provided personalized tutoring for students in grades 9–12 to enhance mathematical understanding and proficiency
- **Improved grades by 15–20%** for multiple students by fostering a supportive, structured learning environment

PROJECTS

MemoLens | *FastAPI, Next.js, PostgreSQL, Redis, WebSockets, Python, Docker*
- Built a **real-time assistive system** for dementia patients and caregivers with live context, reminders, and item tracking
- Engineered a full-stack pipeline: camera input → backend event processing → memory-style queries → real-time WebSocket updates for moment-to-moment caregiver support

Doceo AI-Powered STEM Tutoring | *Next.js, TypeScript, FastAPI, Google Gemini API, SSE*
- Built an interactive AI tutoring product that generates step-by-step STEM lessons, streams voice narration, and animates a whiteboard-style explanation in sync
- Designed Gemini-based prompt pipelines for structured lesson generation and synced narration with animated whiteboard playback via SSE streaming

Data Health Console | *Python, Streamlit, Pandas, NumPy, Matplotlib, SQLite, Pytest*
- Developed a **quantitative data-quality scoring system (0–100)** using penalty-based diagnostics for missing values, duplicates, structural issues, and outliers
- Engineered modular backend architecture with SQLite persistence for dataset history, cleaned snapshots, and generated reports; added automated test coverage with Pytest

OpenClaw Workshop on AWS | *AWS EC2, S3, OpenAI API, FastAPI, Python, Whisper, Discord*
- Designed and delivered a **hands-on cloud workshop** where students build a local AI note-taking app and deploy an OpenClaw agent on AWS
- Guided students through EC2 setup, API integration, and Discord connectivity; kept the setup reproducible with clear model tradeoff explanations

LEADERSHIP & EXTRACURRICULARS

Garth Webb Secondary School Sept 2024 – June 2025
DECA Club Lead Mentor
- Led and mentored **30+ students** in business problem solving, competitive case analysis, and presentation development
- Organized training sessions and mock competitions simulating real-world business scenarios

EDUCATION

McMaster University Sept 2025 – Apr 2030 (Expected)
Candidate for BASc, Computer Science Hamilton, ON, Canada